

MID-1 IMPORTANT QUESTIONS

COMPUTER NETWORKS

UNIT-I :2 MARKS QUESTIONS

1. Define a computer network
2. What are LAN, MAN and WAN?
3. List the different types of network topologies
4. Explain the purpose of the OSI Reference Model.
5. What is the difference between guided and unguided transmission media?

UNIT-I: 5 MARKS QUESTIONS

1. Explain the different types of computer networks: LAN, MAN and WAN with suitable examples.
2. Describe the OSI Reference Model and explain the functions of its layers.
3. Explain the TCP/IP Reference Model and compare it with the OSI Reference Model.
4. Explain guided transmission media with reference to twisted-pair cable, coaxial cable and fiber-optic cable.

UNIT-II :2 MARKS QUESTIONS

1. What is framing in the data link layer?
2. Define flow control.
3. What is CRC?
4. What is the purpose of a checksum in error detection?
5. Explain the basic idea of the Stop-and-Wait protocol.

UNIT-II: 5 MARKS QUESTIONS

1. Explain fixed-size and variable-size framing techniques with suitable examples.
2. Explain error detection and correction techniques used in the data link layer.
3. Explain CRC and Internet checksum methods used for error detection.
4. Explain Sliding Window protocols with reference to One-bit, Go-Back-N and Selective Repeat protocols.
5. Problems on CRC, Check sum, Hamming code.

UNIT-III: 2 Marks QUESTIONS

1. What is random access in the data link layer?
2. Define ALOHA.
3. What is CSMA?
4. Differentiate between CSMA/CD and CSMA/CA.
5. What is channelization? Name the channelization techniques.

UNIT-III: 5 MARKS QUESTIONS

1. Explain the working of ALOHA and discuss its basic operation.
2. Explain CSMA, CSMA/CD and CSMA/CA with suitable examples.
3. Explain controlled access techniques: Reservation, Polling and Token Passing.
4. Explain channelization techniques: FDMA, TDMA and CDMA.